

Solving One-Step Linear Equations

Algebra Worksheet · Grade 6–8

Name: _____

Date: _____

Learning Objectives

- Understand that a linear equation has a highest exponent of one on the variable
- Isolate the variable by applying the inverse operation (addition or subtraction) to both sides of the equation
- Solve one-step linear equations involving positive and negative integers

Problems

1. Solve for x.

$$x + 3 = 7$$

2. Solve for x.

$$x - 5 = 2$$

3. Solve for x.

$$x + 7 = 10$$

4. Solve for x.

$$-5 + x = 2$$

5. Solve for x. Remember: subtracting a larger number from a smaller number gives a negative answer.

$$x + 7 = 2$$

6. Solve for x. Apply your knowledge of adding and subtracting signed numbers.

$$x - 5 = -12$$



7. Solve for x.

$$x + 9 = -4$$

8. Solve for x.

$$x - 8 = -3$$

9. Solve for x.

$$-14 + x = -6$$

10. Solve for x. This problem combines a negative term on the left side with a negative result.

$$x + 15 = -7$$



Solving One-Step Linear Equations — Answer Key

Algebra Worksheet · Grade 6–8

Answer Key

1. Answer: $x = 4$

- Subtract 3 from both sides: $x + 3 - 3 = 7 - 3$
 - $x = 4$
-

2. Answer: $x = 7$

- Add 5 to both sides: $x - 5 + 5 = 2 + 5$
 - $x = 7$
-

3. Answer: $x = 3$

- Subtract 7 from both sides: $x + 7 - 7 = 10 - 7$
 - $x = 3$
-

4. Answer: $x = 7$

- Add 5 to both sides: $-5 + 5 + x = 2 + 5$
 - $x = 7$
-

5. Answer: $x = -5$

- Subtract 7 from both sides: $x + 7 - 7 = 2 - 7$
 - $2 - 7 = -5$, so $x = -5$
-

6. Answer: $x = -7$

- Add 5 to both sides: $x - 5 + 5 = -12 + 5$
 - $-12 + 5 = -7$, so $x = -7$
-

7. Answer: $x = -13$

- Subtract 9 from both sides: $x + 9 - 9 = -4 - 9$
 - $-4 - 9 = -13$, so $x = -13$
-

8. Answer: $x = 5$

- Add 8 to both sides: $x - 8 + 8 = -3 + 8$
 - $-3 + 8 = 5$, so $x = 5$
-

9. Answer: $x = 8$

- Add 14 to both sides: $-14 + 14 + x = -6 + 14$
 - $-6 + 14 = 8$, so $x = 8$
-

10. Answer: $x = -22$





- $-7 - 15 = -22$, so $x = -22$

